FORMULA SHEET / *FORMULEBLAD*

Sample sizes / *Steekproefgroottes*

$$n \geq \frac{z_{\alpha/2}^2 \sigma^2}{e^2},$$

$$n \geq \frac{z_{\alpha/2}^2 \pi(1-\pi)}{e^2},$$

FPC / EPK: $$n \geq \frac{n_0 N}{n_0 + (N-1)}.$$

Inference formulae / *Inferensie formules*

Under certain conditions: / *Onder sekere voorwaardes:*

Parameter	Statistic / *Statistiek*	Standard error / *Standaardfout*	Distribution / *Verdeling*
μ	$\bar{X}$	$\frac{\sigma}{\sqrt{n}}$	$Z \sim N(0,1)$, σ known / *bekend*
μ	$\bar{X}$	$\frac{s}{\sqrt{n}}$	$T \sim t_{n-1}$, σ unknown / *onbekend*
π	p	$\sqrt{\frac{\pi(1-\pi)}{n}}$	$Z \sim N(0,1)$
$\mu_1 - \mu_2$	$\bar{X}_1 - \bar{X}_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$Z \sim N(0,1)$, σ_1, σ_2 known / *bekend*
$\mu_1 - \mu_2$	$\bar{X}_1 - \bar{X}_2$	$\sqrt{s_p^2(\frac{1}{n_1} + \frac{1}{n_2})}$ where/*waar* $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$	$T \sim t_{df=n_1+n_2-2}$, σ_1, σ_2 unknown/*onbekend* ($\sigma_1^2 = \sigma_2^2$)
$\pi_1 - \pi_2$	$p_1 - p_2$	$\sqrt{\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}}$	$Z \sim N(0,1)$
$\pi_1 - \pi_2$	$p_1 - p_2$	$\sqrt{\bar{p}(1-\bar{p})(\frac{1}{n_1} + \frac{1}{n_2})}$	$Z \sim N(0,1)$ $\pi_1 = \pi_2$
$\frac{\sigma_1^2}{\sigma_2^2}$	$\frac{s_1^2}{s_2^2}$		F^{n_1-1, n_2-1}
β_1	b_1	$\frac{s_{YX}}{\sqrt{SSX}}$	$T \sim t_{df=n-2}$
ρ	r	$\sqrt{\frac{1-r^2}{n-2}}$	$T \sim t_{df=n-2}$

Simple linear regression / *Eenvoudige lineêre regressie*

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$SST = SSR + SSE$	$SSX = \sum_{i=1}^{n}(x_i - \bar{x})^2$	$SSXY = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$
$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$	$SSR = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$	$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
$b_1 = \frac{SSXY}{SSX}$	$b_0 = \bar{y} - b_1\bar{x}$	$r^2 = \frac{SSR}{SST}$
$s_{YX} = \sqrt{\frac{SSE}{n-2}}$	$MSR = \frac{SSR}{1}$	$MSE = \frac{SSE}{n-2}$
	$F_{stat} = \frac{MSR}{MSE} \sim F^{1,n-2}$	

Multiple linear regression / *Meervoudige lineêre regressie*

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \varepsilon_i$$

$r^2_{adj} = 1 - [(1 - r^2)(\frac{n-1}{n-k-1})]$
$MSR = \frac{SSR}{k}$
$MSE = \frac{SSE}{n-k-1}$
$F_{stat} = \frac{MSR}{MSE} \sim F^{k,n-k-1}$
$t_{stat} = \frac{b_j - \beta_j}{s_{b_j}}$